

Challenge 8: Ryser’s Hypergraph Conjecture

Definition 1 (r -partite). Given $r \in \mathbb{N}$, an r -uniform hypergraph $\mathcal{H}$ is r -partite if there is a partition

$$V(\mathcal{H}) = \bigcup_{i=1}^r P_i$$

of the vertex set of $\mathcal{H}$ such that $|P_i \cap e| = 1$, for each $e \in E(\mathcal{H})$ and $i \in [r]$.

Definition 2 (Cover number). Given a hypergraph $\mathcal{H}$, a set C of vertices of $\mathcal{H}$ is a *vertex cover* if it contains a vertex of every edge. The smallest cardinality of a vertex cover of $\mathcal{H}$ is called the *cover number* of $\mathcal{H}$. It is denoted by $\tau(\mathcal{H})$.

Definition 3 (Matching number). The *matching number* of a hypergraph $\mathcal{H}$ is the largest cardinality of a set of pairwise disjoint edges of $\mathcal{H}$. It is denoted by $\nu(\mathcal{H})$.

Conjecture 1.1 (Ryser’s Conjecture [1]). Let $r \in \mathbb{N}$ with $r \geq 2$. Every r -partite r -uniform hypergraph $\mathcal{H}$ satisfies:

$$\tau(\mathcal{H}) \leq (r - 1) \cdot \nu(\mathcal{H})$$

For $r = 2$, it is a classical theorem of König, and for $r = 3$ it was proved by Aharoni [2] using the topological method. The hypothesis $r \geq 2$ is necessary: for $r = 1$ the bound reads $\tau(\mathcal{H}) \leq 0$, which already fails for the single-vertex single-edge hypergraph (where $\tau = \nu = 1$).

References

- [1] A. Bishnoi et al. *Ryser’s Conjecture for t -intersecting hypergraphs*. 2020. arXiv: 2001.04132 [math.CO]. URL: <https://arxiv.org/abs/2001.04132>.
- [2] R. Aharoni. “Ryser’s conjecture for tripartite 3-graphs”. In: *Combinatorica* 21.1 (2001), pp. 1–4.